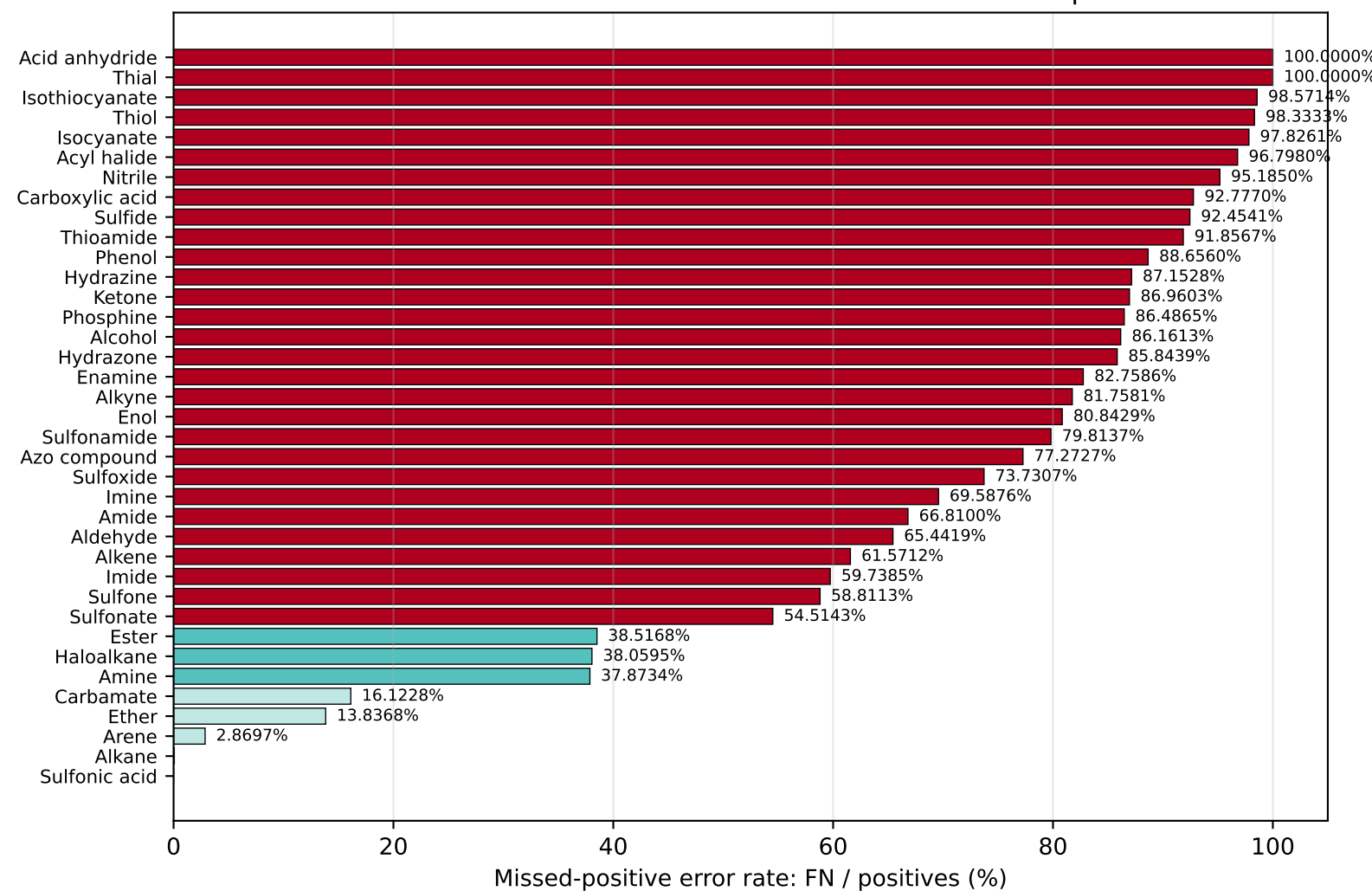
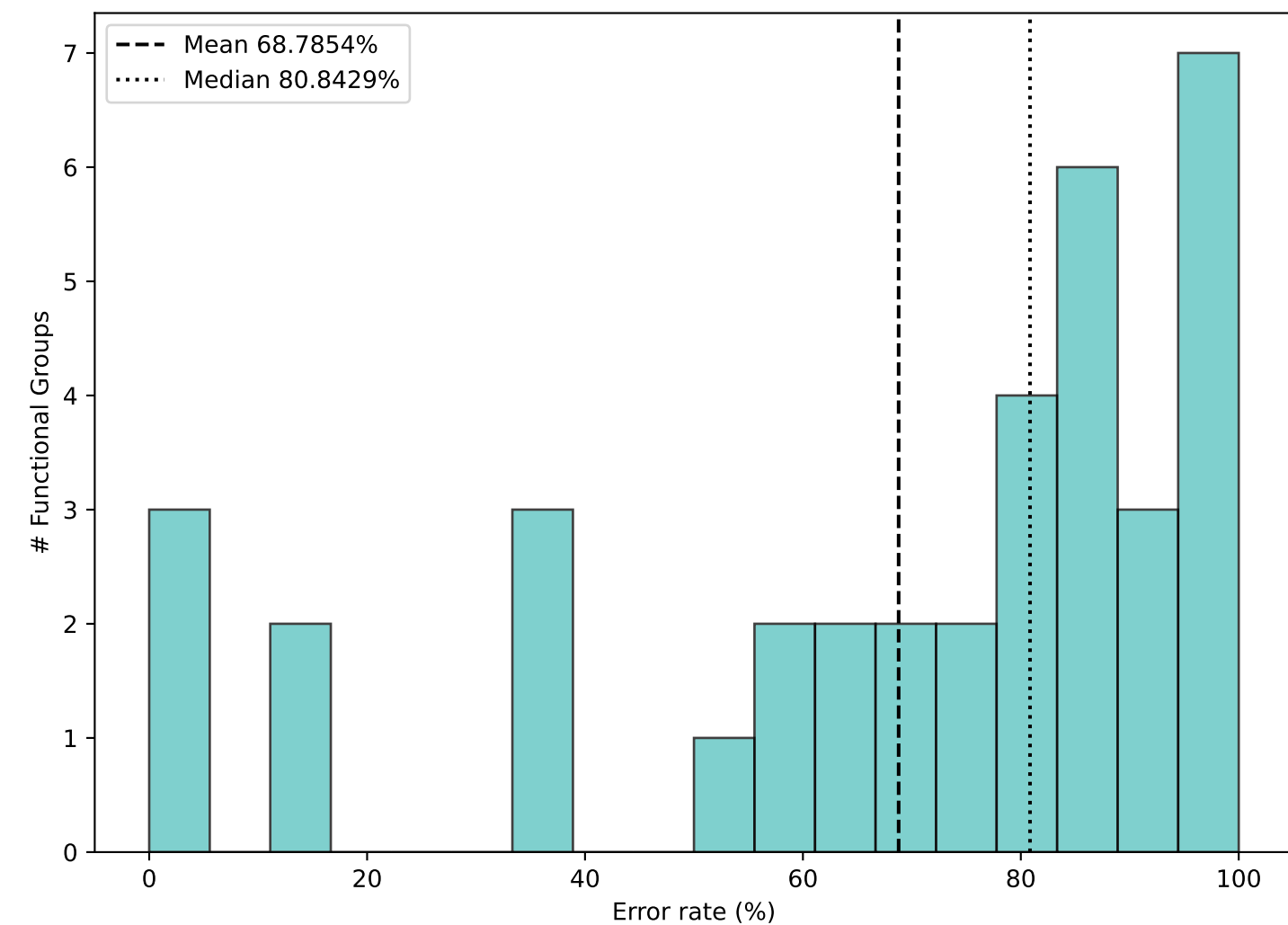


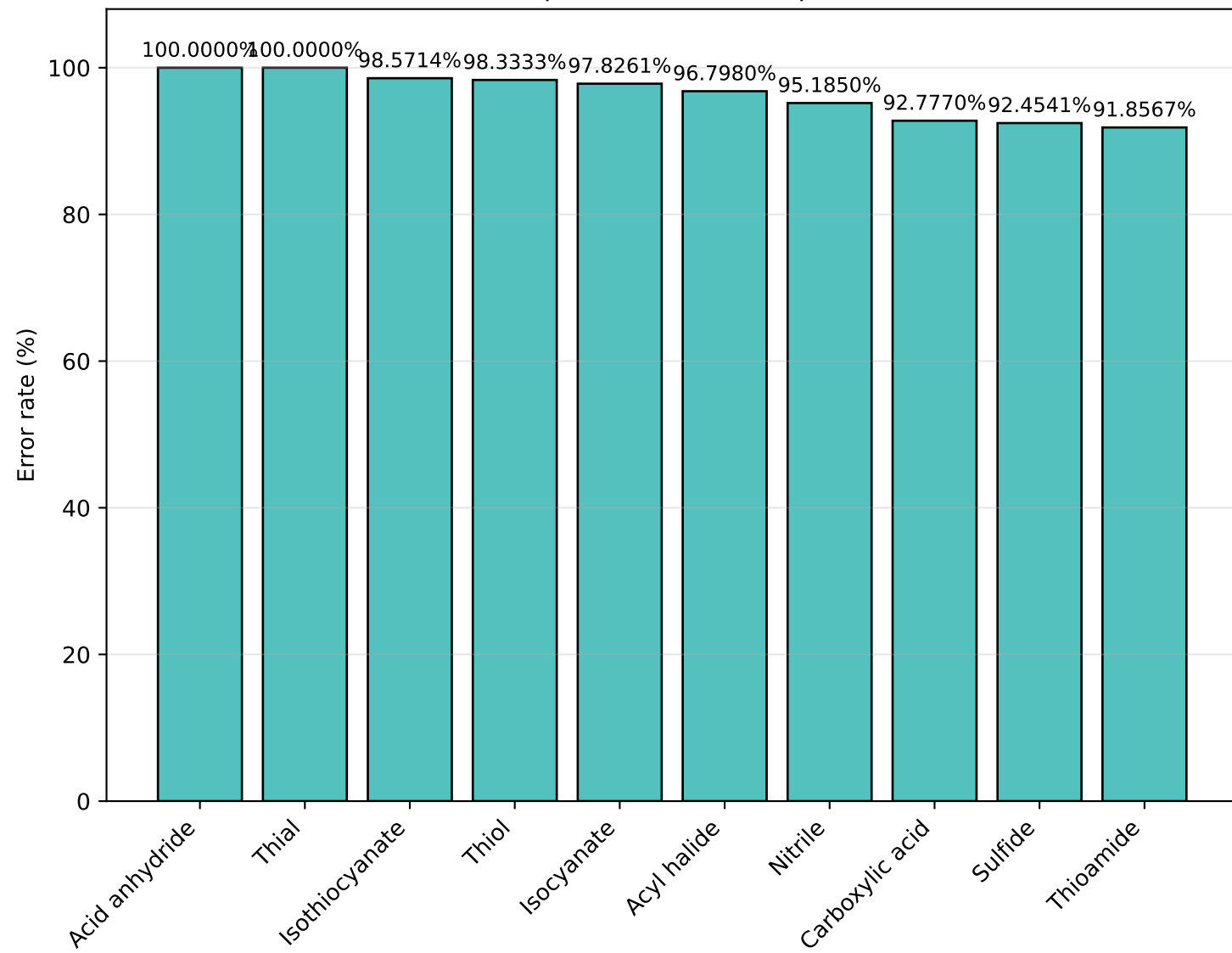
XGBoost H-NMR Test Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



XGB00ST H-NMR TEST SET ERROR ANALYSIS

Global metrics:

F1 micro: 0.7651
F1 macro: 0.3677
Precision micro: 0.8885
Recall micro: 0.6718
Hamming accuracy: 0.9468

Error-rate statistics:

Mean: 68.7854%
Median: 80.8429%
Std: 30.3945%
Max: 100.0000% (Acid anhydride)
Min: 0.0000% (Sulfonic acid)

Problem classes:

Severe (>50%): 29 groups
High (20-50%): 3 groups
Moderate (5-20%): 2 groups
Low (<=5%): 3 groups